Movie Financial Success and Predicting Investment Analytics

Angie Avalos Joel

Tamanna Singh

CS-133 Data Visualization

Dr. Jessica Huynh-Westfall

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**Abstract**

This project explores the relationship between movie attributes and their financial performance using data analysis and machine learning techniques. The dataset includes information such as budget, revenue, popularity, and vote count. Initial exploratory data analysis was conducted using multiple visualizations to identify patterns and trends in the data. Based on these insights, three machine learning models were developed to predict whether a movie would be financially successful. Each model's performance was evaluated using 5-fold cross-validation. The results highlight key factors influencing movie financial success and demonstrate the strengths of different modeling approaches.

**Background**

The film industry is a multi-billion dollar market where investments carry significant financial risk. High budgets do not always guarantee financial success and some films with large budgets fail to return high revenue, while other smaller productions can be profitable.

We investigate the questions:

1. Which movie attributes (budget, popularity, vote count, etc.) have the strongest influence on profitability?
2. How strongly does production budget influence box office revenue?
3. Can we accurately predict whether a movie will be financially successful using machine learning models?
4. Does movie popularity correlate with financial success?
5. Are certain genres more likely to produce profitable films than others?

In this project, we define financial success as profitability, calculated as:

$$Profit = Revenue - Budget$$

A movie is considered financially successful if its revenue exceeds its given budget. Our primary goals in determining success are to analyze how movie features influence financial outcomes, identify key features that contribute to profitable films, explore patterns in movie investments and box office performance, and to build a predictive model that estimates the probability of a movie being profitable.

**Data Exploration and Visualization**

The datasets movies.csv and movies_metadata.csv used for this project were collected from Kaggle.com. The dataset movies.csv includes 4802 rows representing a single movie and 24 columns representing an attribute of that movie (Shinde). It contains variables including, budget, cast, revenue, popularity, and vote average that provide financial and audience insights. The other dataset movies_metadata.csv has similar features and structure allowing us to perform several exploratory data analysis techniques as it contains numerical and categorical variables (Banik). These include analysis between financial and engagement data using scatterplots and bar plots to examine relationships between variables and gain insights for features to include in our machine learning models.

**Data Preparation**

Before initial data exploration and building machine learning models, the dataset was cleaned and preprocessed. Missing values in categorical features were handled by replacing null values with empty strings and a new feature called "main_genre" was created by extracting the first listed genre. Additionally, a new variable called "profit" was calculated as the difference between revenue and budget. For classification modeling, a binary target variable "success" was created where movies with positive profit were labeled as 1. The dataset is split into training and test sets using an 80/20 split and the original dataset with scikit-learn train_test_split. These transformations improve the quality of the dataset and make it suitable for analysis and modeling.

**Model Development**

In this section, we develop machine learning models to predict movie financial success based on features such as budget, popularity, vote count, vote average, and runtime selected because they are related to movie performance and show relationships with financial success during earlier analysis. Three different models were implemented to compare their effectiveness. Logistic Regression was chosen because it models the probability of a movie being successful based on a combination of input features. This is helpful for understanding how individual variables influence the likelihood of success. Decision Tree Classifier was included because it can capture nonlinear relationships between variables. This is useful for movie data where relationships between variables are not purely linear. The last model used is Random Forest Classifier. It is effective for datasets with multiple features and potentially messy data, making it well suited for predicting movie success where many factors contribute to the outcome.

**Evaluation And Testing**

To evaluate the models during training, 5-fold cross-validation was used and their average scores were outputted with scikit- learn. To further evaluate model performance scikit-learn's accuracy_score function was used to provide a measure of how often the model makes correct predictions because the dataset is relatively balanced with 53.8% of movies being classified as successful and 46.2% as unsuccessful. This makes accuracy an appropriate metric for evaluating model performance. The confusion matrix shows how well the model distinguishes between successful and unsuccessful movies and the classification report provides precision, recall, F1-score, and support which provide a deeper look into the model's performance and is appropriate for binary classification.

**Program Design And Organization**

The program is organized as a data analysis and machine learning pipeline. There are four notebooks used that can be found on GitHub: P1_replot-scatter.ipynb, P2_categorical_interactive_transform.ipynb, P3_interactive.ipynb, and machine_learning.ipynb (Avalos Joel and Singh). They begin by importing the necessary libraries, including pandas and Numpy for data processing, matplotlib, seaborn, and Plotly for visualization. For machine learning scikit-learn was used. The program begins by loading the original movie datasets and prepares them for analysis. This includes cleaning the datasets by handling any missing values, removing duplicates, and converting columns to numeric values. The two datasets being used are movies.csv and movies_metadata.csv having them merged through the use of the id column. There was also the creation of various features such as profit, return on investment, success, and main genre. The program then focuses on performing exploratory data analysis. Visualizations such as scatterplots, bar charts, pairplots, and interactive Plotly plots are used to explore relationships between movie attributes such as budget, revenue, popularity, genre, and profit. The interactive scatter plot is able to be filtered based on the genre of the movie demonstrating how revenue changes based on genre with additional information on popularity, budget, and profit as well. The final part of the program focuses on machine learning where dataset splitting and model development and evaluation is done. The best machine learning model is tested using the test set.

**Results**

The results of this project provides insight into the factors that influence movie financial success. A positive relationship between budget and revenue was observed, indicating that films with a larger budget tend to generate higher revenue. This answers our second question with budget having a strong positive influence on revenue.

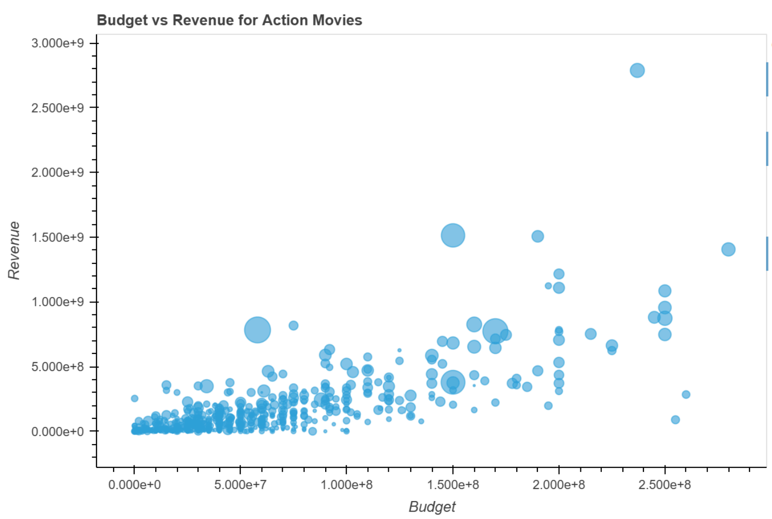


Figure 1: Budget Vs Revenue of Movies

The pairplot shows that popularity and vote count are associated with financial performance, suggesting that audience engagement plays a role in movie success.

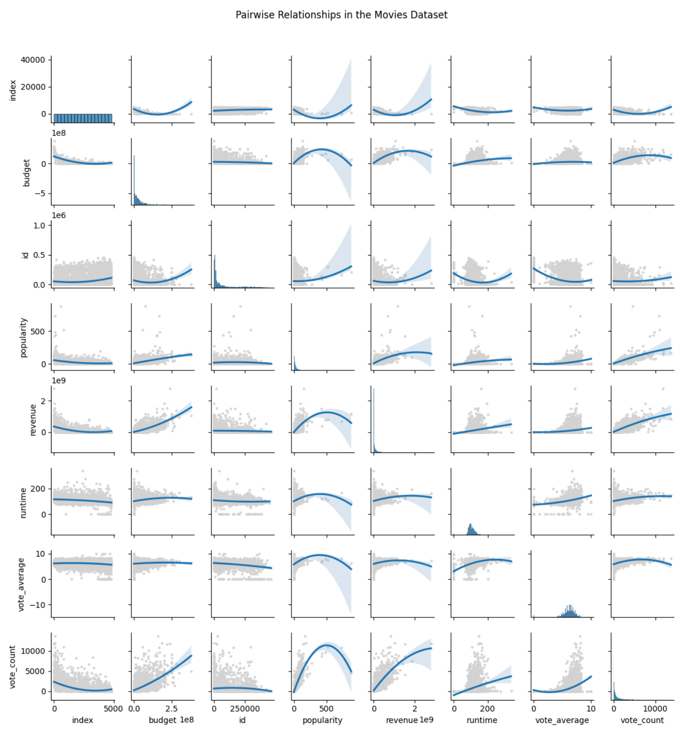


Figure 2: Pairwise Relationships

To answer our fifth question, the analysis of genres indicates that the genres animation, adventure, and family are more likely to produce higher profit compared to others.

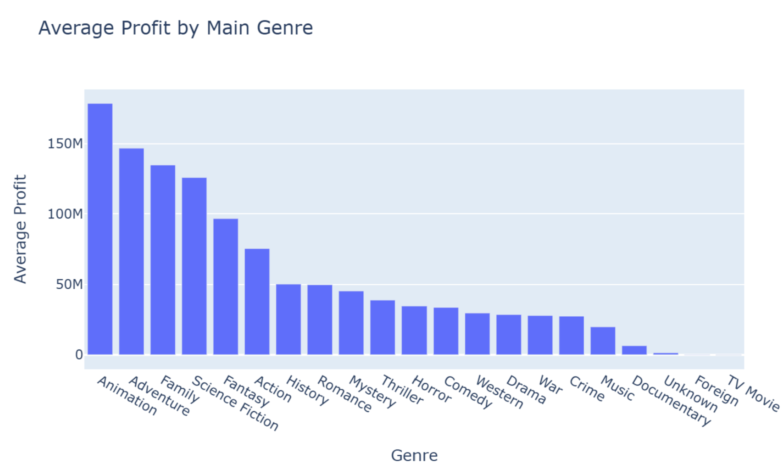


Figure 3: Film's Profit By Genre

Overall, factors such as genre, budget, and audience engagement influence a movie's financial success.

The machine learning models further supported these findings. The dataset was split using scikit-learn's train_test_split function.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

Three models were trained and evaluated.

```
## decision tree
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train_imputed, y_train)
```

```
rf_model = RandomForestClassifier(n_estimators=200, max_depth=10, random_state=42)
rf_model.fit(X_train_imputed, y_train)
```

```
log_model = LogisticRegression(max_iter=1000, random_state=42)
log_model.fit(X_train_scaled, y_train)
```

Cross validation results show the Random Forest model having the highest average score.

```
log_scores = cross_val_score(log_model, X_train_scaled, y_train, cv=5)
```

```
rf_scores = cross_val_score(rf_model, X_train_imputed, y_train, cv=5)
```

```
dt_scores = cross_val_score(dt_model, X_train_imputed, y_train, cv=5)
```

Random forest achieved the highest test accuracy, while Logistic Regression performed slightly worse. The Decision Tree model consistently showed weaker performance.

```
print("\nLogistic Regression Cross-Validation Accuracy:", log_scores.mean())
print("Logistic Regression Test Accuracy:", accuracy_score(y_test, log_prediction))
```

```
print("\nDecision Tree Cross-Validation Accuracy:", dt_scores.mean())
print("Decision Tree Test Accuracy:", accuracy_score(y_test, dt_prediction))
```

```
print("\nRandom Forest Cross-Validation Accuracy:", rf_scores.mean())
print("Random Forest Test Accuracy:", accuracy_score(y_test, rf_prediction))
```

```
Model Comparison:
                 Model  CV Accuracy  Test Accuracy
0  Logistic Regression     0.783333       0.774194
1        Decision Tree     0.781250       0.782518
2        Random Forest     0.789323       0.786681
```

Feature importance analysis using the Random Forest model answers our first question and indicates that vote count, popularity, and budget are the most influential predictors of financial success. No single feature dominated the model suggesting that movie success is influenced by multiple factors.

```
Feature Importance:
              Feature  Importance
3    vote_count_movies    0.391302
1    popularity_movies    0.282417
0        budget_movies    0.165983
2  vote_average_movies    0.083765
4       runtime_movies    0.076533
```

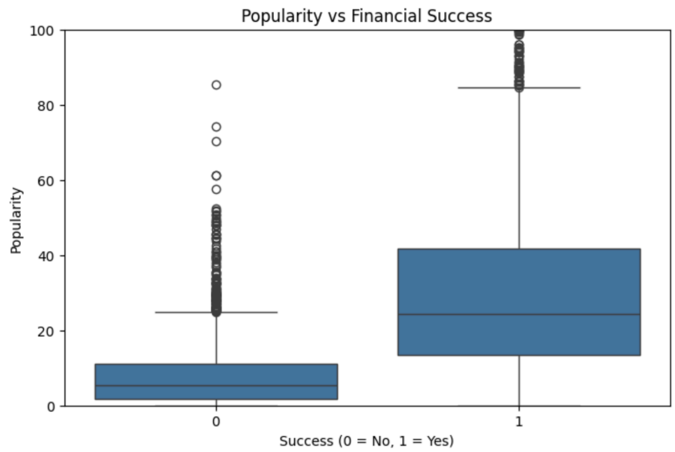


Figure 4: Film's Popularity By Financial Success

We can now answer our fourth question and say that popularity has a positive correlation with financial success. The confusion matrix for Random Forest shows that the model was able to accurately classify if the movie was successful 347 times and unsuccessful 409 times.

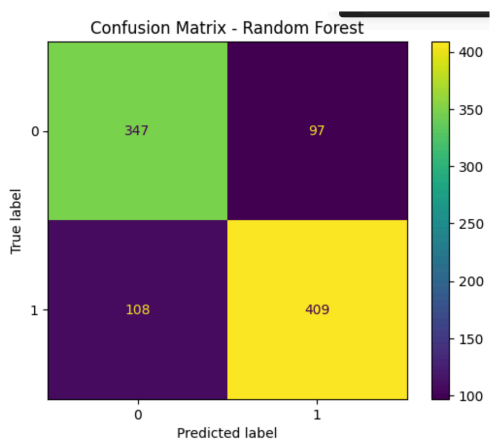


Figure 5: Random Forest Confusion Matrix

To answer our final question, the results demonstrate that while a movie's financial success can be predicted with reasonable accuracy, it is still a complex outcome fueled by interacting financial and audience engagement features.

**Instructions**

To run this project, the user must have Python installed as well as the required libraries. It is recommended to use Jupyter Notebook or Google Colab to run the code. The project files must be downloaded from the github with the datasets of movies.csv and movies_metadata.csv.

1. Open the Jupyter Notebook or upload the notebook to Google Colab
2. Install the following required Python libraries if they are not already installed.
    - Pandas
    - Plotly
    - Numpy
    - Seaborn
    - Scikit-learn
    - Matplotlib
3. The dataset files must be placed in the same directory as the notebook.
4. Run each cell in order from top to bottom.
5. Visualizations and interactive plots created can be viewed after running.
6. The machine learning cells will output model performance including accuracy, confusion matrices, and classification reports.

References

Banik, Rounak. *The Movies Dataset.* Kaggle, 10 Nov. 2017, www.kaggle.com/datasets/rounakbanik/the-movies-dataset?select=movies_metadata.csv.

Shinde, Harsh. *Movies.csv.* Kaggle, 2024, https://www.kaggle.com/datasets/harshshinde8/movies-csv.

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